

CSE2430 - Operating Systems

Homework Threads and Concurrency

Turn-in Instructions

No hand-in required. Questions are meant for self-directed exam preparation and can be discussed in the midterm Q&A session.

Question 1

When executing system or library calls from multi-threaded code, it is important to know which calls are re-entrant and which are not.

- a) What is the meaning of re-entrant?
- b) How does a re-entrant call behave differently from a call that is not re-entrant?
- c) Why is this an important consideration for multi-threaded code?

Question 2

Give a real-life (non-IT) example for a concurrent system. Explain safety and liveness in terms of this system.

Question 3

In the user-level threading model, why does preemption by the operating system (e.g., due to a system call) necessarily preempt the execution of **all** threads?

Question 4

Explain why at some point increasing the number of threads that a program uses often does not increase performance anymore.